Ethereum Address → Identity Expansion

Current Progress Report — Alchemy Proof of Concept

1. Project Objective

Expand the existing dataset of known Ethereum addresses with reputable identity evidence without attempting to enumerate every Ethereum address or process the entire Ethereum node dataset.

Approach: use known addresses as seeds, discover related addresses through Alchemy, then independently check identity evidence for the discovered candidates.

2. Current Alchemy Proof of Concept

Stage	Result
Seed addresses tested	10
Discovery source	Alchemy Ethereum Mainnet API
Unique candidate addresses	4,879
Etherscan verified contracts	316
Verified ENS names	457
Candidates with both sources	47
Candidates with ≥1 identity evidence source	726
Candidates with no identity evidence found	4,153

The 726 figure means that at least one identity-evidence mechanism produced a result. It is not a claim that 726 real-world identities were confirmed: Etherscan contract verification and ENS naming are different evidence types.

3. Pipeline Implemented

Step	Implementation
1. Seed input	Existing Ethereum address dataset.
2. Candidate discovery	Alchemy asset-transfer data around seed addresses.
3. Deduplication	Consolidate discovered addresses into unique candidates.
4. Contract verification	Check Etherscan source-code verification.
5. ENS verification	Reverse ENS name + forward resolution back to the same address.
6. Evidence combination	Combine Etherscan and ENS evidence while keeping sources distinct.

4. Identity Evidence Interpretation

• **Etherscan verified contract:** evidence that contract source code is verified on Etherscan; this is not automatically a real-world person or organization identity.

• **ENS verified name:** an ENS name resolves forward to the same address, providing address-to-name evidence.

• **Multiple sources:** 47 candidates currently have both Etherscan and ENS evidence.

• **No evidence:** 4,153 candidates currently have neither of these two evidence types; this does not prove they have no identity.

5. Current Data Products

File	Purpose
alchemy_relationships.csv	Seed-to-candidate discovery relationships and transfer context.
alchemy_candidates.csv	Unique candidate addresses from the Alchemy experiment.
alchemy_verified_candidates.csv	Etherscan verification results.
alchemy_ens_verified.csv	ENS verification results.
alchemy_identity_evidence.csv	Combined identity-evidence dataset.

6. Important Limitation Before Scaling

The 10-seed run is a proof of concept. Before running the full seed population, the Alchemy discovery script should be strengthened with pagination/pageKey handling, consideration of internal ETH transfers, robust error/retry handling, checkpoint/resume support, and complete discovery provenance.

7. Proposed Next Alchemy-Only Step

Harden the Alchemy discovery pipeline, run a larger controlled test, and measure how many new identity-evidence records are produced as the seed population grows. This keeps the work focused on expanding the identity dataset rather than simply collecting addresses.

8. Meeting Summary

A working Alchemy-based proof of concept is complete: 10 known seed addresses produced 4,879 unique candidates; Etherscan found 316 verified-contract records and ENS produced 457 verified ENS-name records, with 47 supported by both. In total, 726 candidates have at least one of these identity-evidence types. The next step is to improve Alchemy completeness and resumability before scaling.